

October 5, 2025

Q: All words don't end with double letter.

A:  $(a + b)^*(ab + ba)$

Q:  $b^*(abb^*)^*(\Lambda + a)$

A: All words that don't have double a.

Q:  $(\Lambda + a)(ba)^*(\Lambda + b)$

A: All words that don't have double letters.

Q: All words having exactly 2 as or 3 as. Not more or less.

A:  $b^*ab^*ab^* + b^*ab^*ab^*ab^*$

Q:  $(aa + bb + (ab + ba)(aa + bb)^*(ab + ba))^*$

A: All words having even number of as and even number of bs. Also called **even-even** language.

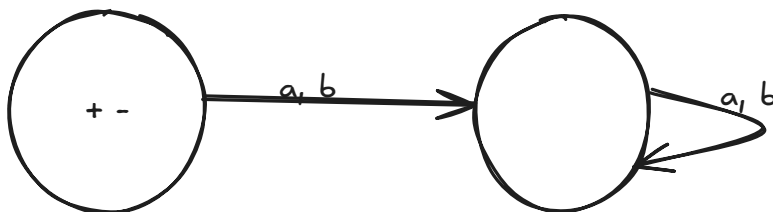
## Finite Automata

Finite number of states, one of which is designated as a start state.

- is start state. + is end state.

Every state **must** have two transitions, one for 'a' another for 'b'.

Finite Automata which accepts only null string



All words  $(a + b)^*$

